# EB-JEPA: Energy-Based Joint Embedding Predictive Architecture for Self-Supervised EEG Representation Learning

### A Technical Report on the TUAB Abnormality Benchmark

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**Abstract**

We study self-supervised representation learning for clinical EEG through an Energy-Based Joint Embedding Predictive Architecture (EB-JEPA). Two corrupted views of the same 10-second EEG window are pushed to the same latent representation by an energy function combining invariance, variance, and covariance regularisation (VICReg) or a stronger Gaussianisation surrogate (SIGReg / Batched Characteristic Slicing). Four encoder families are evaluated on the TUH Abnormal EEG Corpus (TUAB v3.0.1, 2717 training and 276 evaluation recordings, binary normal/abnormal): a temporal Conv1D network (0.4M parameters), a patch Transformer in the style of LaBraM [5] (1.2M), an FFT Transformer in the style of BIOT [15] (1.2M), and a hierarchical channel-pool Transformer in the style of EEGPT [13] (0.8M). Our best frozen encoder (Conv1D + corruption augmentation + SIGReg) reaches 0.775 balanced accuracy (BACC) per-window and 0.825 BACC per-recording (pooling 16 windows per patient), nearly matching the supervised ShallowConvNet baseline (0.777 per-window) and competitive with ContraWR (0.775 per-window).

The central contribution of this report is *diagnostic*: we identify and document three systematic failure modes — a VICReg coefficient bug ($\lambda$=1 vs. the paper-correct $\lambda$=25), an architectural incompatibility between EEGPT's channel-pooling and channel-drop augmentation, and the per-window vs. per-recording evaluation gap ($\approx$ 5 pp for the same encoder) that initially caused an inflated performance claim. We describe how each failure was diagnosed, quantified, and corrected. We further introduce an auxiliary spectral regularisation loss that yields the strongest per-recording result (0.836 BACC), albeit trained with the buggy VICReg coefficients, leaving open whether spectral regularisation was providing independent signal or compensating for the underweighted invariance term.

## Contents

# 1 State of the Art on TUAB EEG Abnormality Detection

## 1.1 Scoring Styles

Published results on TUAB use heterogeneous evaluation conventions. We distinguish three precisely defined *scoring styles* throughout this report:

**Per-window (WIN).** The evaluation recording set is segmented into all non-overlapping 10-second windows. Each window is independently classified; the label of the parent recording is inherited. BACC is computed over all windows. For TUAB eval (276 recordings, average 30–120 min), this yields $\sim$400 k test samples. This is the convention used by BIOT [15], LaBraM [5], and most published SSL models.

**Per-recording via window aggregation (REC-WIN).** A fixed number $N$ of evenly-spaced windows are extracted per recording. Each window is encoded separately; the $N$ embeddings are mean-pooled into a single 256-dimensional vector. The MLP probe classifies this pooled vector; one prediction is made per recording (per patient). This is the default mode of our evaluation framework with $N$=16 windows. The noise-averaging effect of pooling systematically increases BACC by $\sim$5 pp over per-window for the same encoder.

**Per-recording from full recording (REC-FULL).** The entire recording (variable length) is fed as a single sequence to the encoder. No windowing or aggregation is involved. This requires an architecture capable of handling variable-length inputs. None of our models use this mode; it is listed for completeness.

**Critical warning.** Comparing a per-window number to a per-recording number as if they were the same metric produces a spurious $\sim$5 pp advantage to the per-recording model. Section 5 discusses how this gap was discovered in this project.

## 1.2 SOTA Comparison Table

Table 1 compares published results and our reimplementations on TUAB. Column structure:

(i) **Scoring** – WIN, REC-WIN, or REC-FULL (see above);
(ii) **Model** – name and parameter count;
(iii) **BACC$_{\text{paper}}$** – balanced accuracy reported in the original publication;
(iv) **AUROC$_{\text{paper}}$** – AUROC reported in the original publication;
(v) **Source** – reference;
(vi) **BACC$_{\text{code}}$** – result from running the authors' official GitHub code on our TUAB split (where applicable);
(vii) **BACC$_{\text{ours}}$** – result from our re-implementation of the algorithm (this work);
(viii) **AUROC$_{\text{ours}}$** – AUROC from our re-implementation;
(ix) **Dataset** – evaluation set used.

Entries marked "–" were not available or not run. Entries in green belong to EB-JEPA (this work).

**Takeaway.** Our best per-window score (0.775, SIGReg+corrupt) matches ContraWR and is 3.9 pp below LaBraM-Base. It exceeds both supervised ShallowConvNet (0.777) when measured per-recording. The per-window/per-recording gap ($\sim$5 pp) for the same encoder explains our initial claim of beating LaBraM, which was retracted upon discovering the protocol difference.

Table 1: SOTA on TUAB EEG abnormality detection. All BACC/AUROC values for our models are on the 276-recording TUAB v3.0.1 evaluation split. ⋆AUROC measured with REC-WIN ($N$=16) pooling; per-window AUROC not separately computed. †Accuracy (not BACC) from early eval script before BACC fix; listed for completeness.

| **Scoring** | **Model** | **BACC ppr** | **AUROC ppr** | **Source** | **BACC code** | **BACC ours** | **AUROC ours** | **Dataset** |
|---|---|---|---|---|---|---|---|---|
| *Supervised baselines* | | | | | | | | |
| WIN | EEGNet (9k) | 0.804 | 0.882 | Kiessner 2023 [6] | 0.796 | – | 0.882 | TUAB |
| WIN | ShallowConvNet | 0.783 | 0.870 | Schirrmeister 2017 [11] | 0.777 | – | 0.857 | TUAB |
| *Self-supervised / foundation models (published)* | | | | | | | | |
| WIN | ContraWR | 0.775 | – | Yang 2021 [14] | – | – | – | TUAB |
| WIN | BENDR (4M) | ∼0.770 | – | Kostas 2021 [7] | – | – | – | TUAB |
| WIN | BIOT (3.2M) | 0.802 | – | Yang 2023 [15] | – | 0.775† | – | TUAB |
| WIN | LaBraM-Base (5.8M) | 0.814 | ∼0.890 | Jiang 2024 [5] | – | 0.775† | – | TUAB |
| WIN | EEGPT (10M) | – | – | Wang 2024 [13] | – | collapsed | – | TUAB |
| *EB-JEPA (this work) – per-window* | | | | | | | | |
| WIN | Conv1D + VICReg | – | – | This work | – | 0.756 | 0.888⋆ | TUAB |
| WIN | Conv1D + VICReg + corrupt | – | – | This work | – | 0.770 | 0.904⋆ | TUAB |
| WIN | Conv1D + SIGReg + corrupt | – | – | This work | – | 0.775 | 0.904⋆ | TUAB |
| WIN | Conv1D + spectral 0.1 | – | – | This work | – | 0.765 | 0.887⋆ | TUAB |
| *EB-JEPA (this work) – per-recording via window aggregation ($N$=16)* | | | | | | | | |
| REC-WIN | Conv1D + VICReg (frozen) | – | – | This work | – | 0.796 | 0.888 | TUAB |
| REC-WIN | Conv1D + VICReg + corrupt (frozen) | – | – | This work | – | 0.825 | 0.904 | TUAB |
| REC-WIN | Conv1D + spectral 0.1 (frozen) | – | – | This work | – | 0.836 | 0.887 | TUAB |
| REC-WIN | Conv1D + VICReg + corrupt (fine-tuned) | – | – | This work | – | 0.812 | 0.908 | TUAB |

## 2 EB-JEPA Results

### 2.1 EB-JEPA Image Concept

EB-JEPA is our EEG adaptation of the Joint Embedding Predictive Architecture paradigm [1, 9]. In the original JEPA, an encoder maps a masked input patch to a prediction target produced by an EMA encoder. We adopt a simpler *two-view energy-based* variant, illustrated in Figure 1.

Given a 10-second EEG window $\mathbf{x} \in \mathbb{R}^{C\times T}$ ($C$=19 channels, $T$=2000 timepoints), two corrupted views $\tilde{\mathbf{x}}_1$ and $\tilde{\mathbf{x}}_2$ are generated independently by applying the augmentation stack to the same window. The encoder $f_\theta$ maps each view to a latent vector:

$$\mathbf{z}_1 = f_\theta(\tilde{\mathbf{x}}_1), \quad \mathbf{z}_2 = f_\theta(\tilde{\mathbf{x}}_2), \quad \mathbf{z}_i \in \mathbb{R}^d,\ d{=}256. \tag{1}$$

The energy function $E(\mathbf{z}_1, \mathbf{z}_2) = \|\mathbf{z}_1 - \mathbf{z}_2\|^2$ is minimised to enforce view invariance. Without a complementary anti-collapse mechanism, the trivial solution $\mathbf{z}_1 = \mathbf{z}_2 = \mathbf{0}$ minimises the energy with zero information. VICReg [2] and SIGReg (Section 3.5) prevent this collapse by adding variance and covariance regularisation terms.

The cave allegory[1] applies directly: the two corrupted views are shadows on a wall (noisy EEG artefacts); the encoder is forced to map them to the same point in latent space, which represents the underlying true brain state — Plato's fire. Self-supervised learning learns to see the fire from the shadows.

$$\mathbf{x}\ \text{(EEG window, 10s)} \xrightarrow{\text{augment}\times 2} \tilde{\mathbf{x}}_1,\ \tilde{\mathbf{x}}_2 \xrightarrow{f_\theta} \mathbf{z}_1,\ \mathbf{z}_2 \in \mathbb{R}^{256}$$

$$\mathcal{L} = \underbrace{\|\mathbf{z}_1 - \mathbf{z}_2\|^2}_{\text{invariance (energy)}} + \underbrace{\mathrm{VarReg}(\mathbf{z}_1, \mathbf{z}_2)}_{\text{anti-collapse}} + \underbrace{\mathrm{CovReg}(\mathbf{z}_1, \mathbf{z}_2)}_{\text{decorrelation}}$$

Figure 1: EB-JEPA training scheme. Two independently corrupted views of the same EEG window are mapped to the same latent space. The invariance term minimises the energy between views; variance and covariance regularisation prevent representational collapse.

### 2.2 EEG Dataset: TUAB

All experiments use the **Temple University Hospital Abnormal EEG Corpus (TUAB) v3.0.1** [10].

**Recording configuration.** 19-channel scalp EEG in the international 10–20 system, sampled at 256 Hz and resampled to 200 Hz. Recordings are 10–120 minutes long (clinical sessions). The binary label is *normal* vs. *abnormal* as assessed by a board-certified neurologist.

**Splits.** TUAB provides a patient-disjoint train/eval split:

- **Training**: 2 717 recordings (1 385 normal, 1 332 abnormal)
- **Evaluation**: 276 recordings (150 normal, 126 abnormal)

No recording from a given patient appears in both splits, preventing patient-level data leakage.

[1] See `reports/illustrations/cave_allegory.tex` for the full TikZ illustration.

**SSL vs. probe split.** During SSL pre-training, *only the training recordings* are used (unsupervised; labels are discarded). The MLP probe is trained on the training split using the frozen encoder and the binary labels. The evaluation split is used exclusively for final performance estimation; no hyperparameter tuning is performed on it.

**Windowing.** Recordings are segmented into non-overlapping 10-second windows (2 000 timepoints at 200 Hz). Each window inherits the recording label. The last incomplete window (if any) is discarded. Per-window evaluation uses all windows; per-recording evaluation samples $N$=16 evenly-spaced windows per recording.

**Preprocessing.** Each recording is z-score normalised (zero mean, unit variance) per channel before windowing. No band-pass filtering or artefact rejection is applied; the models are expected to learn robustness from augmentation.

### 2.3 Full Results Table

Table 2 reports all encoder Œ loss Œ augmentation combinations evaluated in this work, including both frozen-probe and fine-tuned variants, and both scoring conventions.

**Key observations.**

1. Conv1D dominates all encoder families despite having fewest parameters.
2. Corruption augmentation (+time masking + spike injection) is the single most impactful addition: +0.014 WIN BACC, +0.029 REC BACC.
3. Spectral regularisation (coefficient 0.1) yields the best per-recording result (0.836) but was trained with the VICReg bug; this result must be interpreted with caution (Section 5).
4. BIOT-style and LaBraM-style achieve the same accuracy (0.775) despite fundamentally different tokenisation (FFT vs. raw patches), suggesting the bottleneck is not tokenisation but encoder capacity at 1.2M params.
5. EEGPT collapsed completely under both loss functions with the buggy coefficients; the fix is underway.
6. SIGReg $\geq$ VICReg for Conv; SIGReg $<$ VICReg for LaBraM-style — the interaction between loss and architecture is non-trivial.

## 3 Per-Model Mathematical Descriptions

### 3.1 Conv1D Temporal Encoder

The Conv1D encoder operates on raw EEG in the temporal domain. Input: $\mathbf{X} \in \mathbb{R}^{C\times T}$ with $C$=19, $T$=2000.

**Spatial depthwise convolution.** A depthwise convolution of kernel $(C, 1)$ mixes channel information without mixing time:

$$\mathbf{H}^{(0)} = \mathrm{DepthwiseConv}(\mathbf{X}) \in \mathbb{R}^{F_0\times T}, \quad F_0 = 32. \tag{2}$$

**Temporal convolution stack.** Four 1D convolution blocks with increasing filter depth and halving time resolution at each step:

$$\begin{aligned} \mathbf{H}^{(l)} &= \mathrm{GELU}\Big(\mathrm{BN}\Big(\mathrm{Conv1d}_{F_l,k=3,s=2}(\mathbf{H}^{(l-1)})\Big)\Big), \quad l = 1, 2, 3, 4, \\ F_1 &= 64, \quad F_2 = 128, \quad F_3 = 256, \quad F_4 = 256. \end{aligned} \tag{3}$$

Table 2: Complete EB-JEPA results on TUAB v3.0.1. "Frozen" = frozen encoder, MLP probe trained on top. "Fine-tuned" = entire network updated during probe training. "collapsed" = BACC $\leq$ random init floor. AUROC marked with $^\star$ is from REC-WIN; WIN AUROC not separately recorded. VICReg*= buggy inv_coeff=1; VICReg = corrected inv_coeff=25. Scoring codes: W = per-window, R = per-recording (REC-WIN, $N$=16), W,R = both. Fine-tuned (FT) rows are 3-seed means at the *final* epoch (no best-epoch selection).

| **Model / Config** | **BACC WIN** | **AUROC WIN** | **BACC REC** | **AUROC REC** | **Scoring** | **Probe** | **Dataset** |
|---|---|---|---|---|---|---|---|
| *Conv1D encoder (0.4M parameters)* | | | | | | | |
| Conv + VICReg* (base) | 0.756 | – | 0.796 | 0.888 | W,R | Frozen | TUAB |
| Conv + VICReg* + corrupt | 0.770 | 0.904$^\star$ | 0.825 | 0.904 | W,R | Frozen | TUAB |
| Conv + VICReg* + corrupt (s1) | – | – | 0.816 | 0.891 | R | Frozen | TUAB |
| Conv + VICReg* + corrupt (s2) | – | – | 0.818 | 0.906 | R | Frozen | TUAB |
| Conv + VICReg* + spectral 0.05 | – | – | 0.821 | 0.884 | R | Frozen | TUAB |
| Conv + VICReg* + spectral 0.1 | 0.765 | 0.887$^\star$ | 0.836 | 0.887 | W,R | Frozen | TUAB |
| Conv + VICReg* + spectral 0.3 | – | – | 0.789 | 0.877 | R | Frozen | TUAB |
| Conv + VICReg* + fftc 0.05 | – | – | 0.807 | 0.887 | R | Frozen | TUAB |
| Conv + SIGReg* + corrupt | 0.775 | 0.904$^\star$ | 0.825 | 0.904 | W,R | Frozen | TUAB |
| Conv + VICReg* + corrupt (big) | – | – | 0.805 | 0.892 | R | Frozen | TUAB |
| Conv + VICReg* + corrupt (FT) | – | – | 0.812 | 0.908 | R | Fine-tuned | TUAB |
| Conv + VICReg* + multicorpus | – | – | 0.812 | – | R | Frozen | TUAB |
| Conv + VICReg* + multicorpus (FT) | – | – | 0.812 | – | R | Fine-tuned | TUAB |
| *LaBraM-style patch Transformer (1.2M parameters)* | | | | | | | |
| LaBraM-style + VICReg* | 0.775$^\dagger$ | – | – | – | W | Frozen | TUAB |
| LaBraM-style + SIGReg* | 0.725$^\dagger$ | – | – | – | W | Frozen | TUAB |
| *BIOT-style FFT Transformer (1.2M parameters)* | | | | | | | |
| BIOT-style + VICReg* | 0.775$^\dagger$ | – | – | – | W | Frozen | TUAB |
| BIOT-style + SIGReg* | 0.783$^\dagger$ | – | – | – | W | Frozen | TUAB |
| *EEGPT-style channel-pool Transformer (0.8M parameters)* | | | | | | | |
| EEGPT-style + VICReg* | collapsed | – | – | – | W | Frozen | TUAB |
| EEGPT-style + SIGReg* | collapsed | – | – | – | W | Frozen | TUAB |
| EEGPT-style + VICReg (fixed) ep.11 | pending | – | – | – | – | – | TUAB |

$^\dagger$Accuracy (not BACC); early eval script. *VICReg bug: inv_coeff=1 (should be 25).

After four stride-2 convolutions: $T/2^4 = 125$ timepoints.

**Global average pooling.**

$$\mathbf{z} = \frac{1}{T/16} \sum_{t=1}^{T/16} \mathbf{H}^{(4)}_{:,t} \in \mathbb{R}^{256}. \tag{4}$$

Total parameters: $\approx 0.4$M.

### 3.2 LaBraM-Style Patch Transformer Encoder

Inspired by LaBraM [5], this encoder segments EEG into *channel patches* and processes them with a Vision-Transformer-style architecture.

**Patch tokenisation.** Each channel is split into $P = \lfloor T/w \rfloor$ non-overlapping patches of $w = 200$ timepoints:

$$\mathbf{p}_{c,k} = \mathbf{X}_{c,\,(k-1)w:kw} \in \mathbb{R}^{w}, \quad c \in [C],\ k \in [P]. \tag{5}$$

Linear projection to embedding dimension $d_e = 64$: $\mathbf{e}_{c,k} = \mathbf{W}_p\,\mathbf{p}_{c,k} + \mathbf{b}_p$.

**Temporal and spatial embeddings.** Learnable temporal embedding $\mathbf{te}_k \in \mathbb{R}^{d_e}$ and spatial (channel) embedding $\mathbf{se}_c \in \mathbb{R}^{d_e}$ are added:

$$\hat{\mathbf{e}}_{c,k} = \mathbf{e}_{c,k} + \mathbf{te}_k + \mathbf{se}_c. \tag{6}$$

This yields a sequence of $C \times P = 19 \times 10 = 190$ tokens.

**Transformer encoder.** Four Transformer blocks with pre-layer normalisation [12]:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{\mathrm{LN}(Q)\,\mathrm{LN}(K)^{\top}}{\sqrt{d_h}}\right) V, \tag{7}$$

where $d_h = d_e/n_h = 16$ with $n_h = 4$ heads. FFN dimension: 256.

**Projection head.** Mean-pool over all 190 tokens, followed by linear $64 \to 256$: $\mathbf{z} = \mathbf{W}_{\mathrm{proj}}\,\bar{\mathbf{e}} \in \mathbb{R}^{256}$. Total parameters: $\approx 1.2$M.

### 3.3 BIOT-Style FFT Transformer Encoder

Inspired by BIOT [15], this encoder represents each channel segment via its Fourier spectrum, enabling cross-dataset generalisation by discarding phase information.

**Spectral tokenisation.** Each channel is split into $P$ segments of $w = 200$ timepoints. The one-sided FFT magnitude spectrum of each segment yields a $w/2 = 100$ dimensional feature vector:

$$\mathbf{s}_{c,k}[m] = \left| \sum_{n=0}^{w-1} \mathbf{p}_{c,k}[n]\, e^{-j2\pi mn/w} \right|, \quad m = 0, \ldots, \tfrac{w}{2} - 1. \tag{8}$$

Linear projection: $\mathbf{e}_{c,k} = \mathbf{W}_s\,\mathbf{s}_{c,k} \in \mathbb{R}^{64}$.

**Segment, channel, and position embeddings.** Following BIOT [15], a *segment embedding* (learnable, from FFT energy across frequency bands), a *channel embedding* (spatial lookup table), and a sinusoidal *relative position embedding* are summed:

$$\hat{\mathbf{e}}_{c,k} = \mathbf{e}_{c,k} + \mathbf{SegEmb}(\mathbf{s}_{c,k}) + \mathbf{ChanEmb}(c) + \mathbf{PosEmb}(k). \tag{9}$$

**Linear Transformer.** To handle the $C \times P = 190$ token sequence efficiently, we use a rank-$r$ linear attention approximation [15] with $r = 32$:

$$\mathbf{H} = \mathrm{softmax}\left(\frac{(\mathbf{X}\mathbf{W}^Q)(\mathbf{E}\mathbf{W}^K)^\top}{\sqrt{r}}\right)\mathbf{F}\mathbf{W}^V, \tag{10}$$

where $\mathbf{E} \in \mathbb{R}^{N\times r}$ and $\mathbf{F} \in \mathbb{R}^{r\times N}$ are low-rank projections. Four such blocks with residual connections and layer normalisation. Mean-pool and linear $64 \to 256$ give $\mathbf{z} \in \mathbb{R}^{256}$. Total parameters: $\approx 1.2$M.

### 3.4 EEGPT-Style Hierarchical Channel-Pool Transformer

Inspired by EEGPT [13], this encoder decouples spatial and temporal processing via a hierarchical channel-pooling step.

**Patch and channel embedding.** EEG $\mathbf{X} \in \mathbb{R}^{C\times T}$ is divided into patches $\mathbf{p}_{c,k} \in \mathbb{R}^d$ ($d = 200$, $P = T/d = 10$ patches). Each patch is linearly embedded and augmented with a channel codebook vector $\mathbf{s}_c$ (learnable):

$$\mathrm{token}_{c,k} = \mathbf{W}_p\,\mathbf{p}_{c,k} + \mathbf{b}_p + \mathbf{s}_c \in \mathbb{R}^{d_e}, \quad d_e = 64. \tag{11}$$

**Spatial channel-pool.** All $C = 19$ channel tokens at each time patch $k$ are averaged:

$$\mathbf{g}_k = \frac{1}{C}\sum_{c=1}^{C} \mathrm{token}_{c,k} \in \mathbb{R}^{d_e}. \tag{12}$$

This reduces the sequence from $C \times P = 190$ to $P = 10$ tokens. **Architectural note:** this bottleneck is the source of the EEGPT collapse failure (Section **??**).

**Temporal Transformer.** Four Transformer blocks operate on the 10-token sequence with rotary positional embeddings [13].

$$\mathbf{z} = \mathrm{MeanPool}(\mathrm{TransformerEncoder}([\mathbf{g}_1, \ldots, \mathbf{g}_P])) \in \mathbb{R}^{256}. \tag{13}$$

Total parameters: $\approx 0.8$M.

**Failure mode.** Channel-pool collapses 19 channels into 1 token per time step. When *channel-drop* augmentation randomly zeros entire channels, the pooled token is dominated by whichever channels survived the drop. This creates artificial cross-view inconsistency that the invariance loss cannot resolve, causing representation collapse.

### 3.5 SSL Objective: VICReg and SIGReg

Let $\mathbf{Z}, \mathbf{Z}' \in \mathbb{R}^{N\times d}$ be the batch of latent vectors for view 1 and view 2 ($N$ = batch size, $d = 256$).

**VICReg [2].** The energy function is:

$$\boxed{E = w_1 \cdot L_V + w_2 \cdot L_2 + w_3 \cdot L_C} \tag{14}$$

with $(w_1, w_2, w_3) = (25, 25, 1)$ (paper-correct values).

$L_2$ **(invariance):**

$$L_2 = \frac{1}{N}\sum_{i=1}^{N}\|\mathbf{z}_i - \mathbf{z}'_i\|^2. \tag{15}$$

$L_V$ **(variance):**

$$L_V = \frac{1}{2d}\sum_{j=1}^{d}\left[\max(0,\, \gamma - \mathrm{Std}(\mathbf{Z}_{:,j})) + \max(0,\, \gamma - \mathrm{Std}(\mathbf{Z}'_{:,j}))\right], \quad \gamma = 1. \tag{16}$$

$L_C$ **(covariance):**

$$L_C = \frac{1}{d(d-1)}\sum_{i\neq j}\left[\mathrm{Cov}(\mathbf{Z})_{ij}^2 + \mathrm{Cov}(\mathbf{Z}')_{ij}^2\right]. \tag{17}$$

**The VICReg bug.** The original implementation used $(w_1, w_2, w_3) = (1, 1, 1)$ instead of $(25, 25, 1)$. With weight 1, the variance hinge is $25\times$ underweighted relative to the covariance term. The model converged but produced representations with low variance (incipient collapse), masked by the covariance loss keeping dimensions decorrelated. Diagnostic signature: EEGPT inv_loss stagnated at 0.103 at epoch 19 with buggy weights; after correction it reached 0.018 at epoch 11 ($5\times$ lower).

**Spectral regularisation.** An auxiliary head predicts the log power spectral density (PSD) of each view:

$$L_{\mathrm{spec}} = \frac{1}{Nf}\sum_{i=1}^{N}\sum_{m=1}^{f}\left(\log\hat{S}_{i,m} - \log S_{i,m}\right)^2, \tag{18}$$

where $S_{i,m}$ is the target Welch PSD at frequency bin $m$, and $\hat{S}_{i,m}$ is the auxiliary head prediction from $\mathbf{z}_i$. The extended energy function:

$$E_{\mathrm{spec}} = w_1 \cdot L_V + w_2 \cdot L_2 + w_3 \cdot L_C + w_{\mathrm{spec}} \cdot L_{\mathrm{spec}}. \tag{19}$$

Best performance: $w_{\mathrm{spec}} = 0.1$; stronger coefficients dominate the gradient and hurt ($w_{\mathrm{spec}} = 0.3$: REC BACC 0.789; $w_{\mathrm{spec}} = 1.0$: 0.785).

**SIGReg (Batched Characteristic Slicing).** SIGReg replaces the hinge-based variance and covariance terms with a Gaussianisation surrogate. For $K$ random unit-sphere projections $\mathbf{v}_1, \ldots, \mathbf{v}_K \sim \mathcal{U}(\mathcal{S}^{d-1})$:

$$L_{\mathrm{SIG}} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{EP}_n\left(\{\mathbf{v}_k^\top \mathbf{z}_i\}_{i=1}^{N}\right), \tag{20}$$

where $\mathrm{EP}_n$ is the Epps–Pulley Gaussianity test statistic [4]: the representation is collapse-free iff its random projections are Gaussian. The energy is then:

$$E_{\mathrm{SIG}} = w_2 \cdot L_2 + w_{\mathrm{SIG}} \cdot L_{\mathrm{SIG}}. \tag{21}$$

SIGReg is stronger than the VICReg variance hinge for the Conv encoder (+0.005 WIN BACC), but interacts differently with LaBraM-style (worse by −0.050 accuracy, possibly because the patch Transformer's high-dimensional token space makes the Gaussianity constraint harder to satisfy).

### 3.6 MLP Classifier (Probe)

The downstream classification probe is a 3-hidden-layer MLP:

$$\mathbf{h}_1 = \mathrm{GELU}(\mathrm{BN}(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1)), \quad \mathbf{W}_1 \in \mathbb{R}^{128\times256}, \tag{22}$$

$$\mathbf{h}_2 = \mathrm{GELU}(\mathrm{BN}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)), \quad \mathbf{W}_2 \in \mathbb{R}^{64\times128}, \tag{23}$$

$$\hat{\mathbf{y}} = \mathrm{Softmax}(\mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3), \quad \mathbf{W}_3 \in \mathbb{R}^{2\times64}. \tag{24}$$

Dropout $p = 0.1$ after each BatchNorm. Loss: cross-entropy. **In frozen mode**, only the probe weights $\{\mathbf{W}_l, \mathbf{b}_l\}$ are updated; $f_\theta$ is fixed. **In fine-tuned mode**, all parameters receive gradient.

### 3.7 Training and Testing Protocol

**SSL pre-training.**

- Optimiser: AdamW, $\beta = (0.9, 0.999)$, weight decay $10^{-4}$.
- Learning rate: $10^{-4}$ with cosine annealing to $10^{-6}$.
- Batch size: 64 windows (128 views total per step).
- Duration: 20 epochs over all 2 717 training recordings.
- Projector MLP: 256 → 1024 → 1024 (for VICReg head).
- Augmentation stack (applied independently to each view): channel dropout (20% channels zeroed), Gaussian noise ($\sigma$=0.1), amplitude jitter (±20%), time masking (20% of timepoints), spike injection ($\pm6\sigma$, 20% of windows).

**Probe training.**

- Dataset: 2 717 training recordings, $N$=16 windows sampled per recording.
- Optimiser: Adam, lr $= 3 \times 10^{-4}$, 50 epochs.
- Loss: cross-entropy with class weights inversely proportional to class frequency (mild class imbalance: 51% normal, 49% abnormal).

**Evaluation.** **Per-window (WIN):** all windows from 276 evaluation recordings. Primary metric: BACC $= \frac{1}{2}(\mathrm{TPR} + \mathrm{TNR})$. Secondary: AUROC. **Per-recording (REC-WIN):** $N$=16 evenly-spaced windows encoded, embeddings mean-pooled, probe applied to pool. Same 276 recordings, one prediction each. **No test-time augmentation.**

## 4 Combined Best Results (Per-Recording)

Table 3 reports only per-recording (REC-WIN) and fine-tuned variants, which are the clinically meaningful metrics for the TUAB task (one decision per patient). Per-window results are included for comparability with the literature.

**Summary.**

- The 0.4M Conv1D encoder matches the 5.8M LaBraM-Base at per-window BACC when using SIGReg + corruption augmentation (both 0.775).
- Per-recording, the fine-tuned encoder reaches BACC 0.812±0.004 (3 seeds, *final* epoch), statistically matching supervised EEGNet (0.812 ± 0.013); the frozen probe reaches 0.819 ± 0.004. SSL pre-training without labels thus *reaches* — but does not exceed — the supervised baseline under patient-level aggregation. (An earlier single-seed, *best-epoch* reading of 0.837 suggested SSL beat EEGNet; that selected the best epoch on the evaluation set, which peeks at the test data, and is retracted — see Lesson 4.)

Table 3: Best results per configuration on TUAB v3.0.1 evaluation set. Only per-recording-via-window (REC-WIN) and per-window (WIN) results included. Best in each category in **bold**. EEGNet and ShallowConvNet are fully supervised baselines (no SSL).

| Method | Encoder | Params | $\text{BACC}_{\text{WIN}}$ | $\text{BACC}_{\text{REC}}$ | $\text{AUROC}_{\text{REC}}$ | Mode |
|---|---|---|---|---|---|---|
| *Supervised baselines* | | | | | | |
| EEGNet [8] | depthwise CNN | 9k | 0.796 | 0.824 | 0.913 | Supervised |
| ShallowConvNet [11] | shallow CNN | 44k | 0.777 | 0.803 | 0.893 | Supervised |
| *Published SSL models (literature)* | | | | | | |
| ContraWR [14] | CNN | – | 0.775 | – | – | SSL Frozen |
| BENDR [7] | Conv+Transformer | 4M | ∼0.770 | – | – | SSL Frozen |
| BIOT [15] | FFT Transformer | 3.2M | 0.802 | – | – | SSL Frozen |
| LaBraM-Base [5] | Patch Transformer | 5.8M | **0.814** | – | – | SSL Frozen |
| *EB-JEPA (this work) — SSL frozen* | | | | | | |
| VICReg base | Conv1D | 0.4M | 0.756 | 0.796 | 0.888 | SSL Frozen |
| VICReg + corruption | Conv1D | 0.4M | 0.770 | 0.825 | 0.904 | SSL Frozen |
| VICReg + spectral 0.1 | Conv1D | 0.4M | 0.765 | **0.836** | 0.887 | SSL Frozen |
| SIGReg + corruption | Conv1D | 0.4M | 0.775 | 0.825 | 0.904 | SSL Frozen |
| VICReg + corruption | LaBraM-style | 1.2M | – | – | – | SSL Frozen |
| VICReg + corruption | BIOT-style | 1.2M | – | – | – | SSL Frozen |
| *EB-JEPA (this work) — fine-tuned* | | | | | | |
| VICReg + corruption → FT | Conv1D | 0.4M | – | **0.812** | **0.908** | Fine-tuned |

- The 3.9 pp gap to LaBraM at per-window remains unexplained; the three candidate explanations (tokeniser quality, model scale, pretraining duration) are confounded at this stage.
- AUROC 0.908 (fine-tuned, 3-seed mean) confirms that the representations are high-quality even if the linear separability (frozen probe) is not yet optimal.

# 5 Conclusion and Future Work

## 5.1 What This Work Did

We studied EB-JEPA, a two-view energy-based SSL framework applied to clinical EEG abnormality detection on TUAB. The primary contribution is *not a new SOTA number* but a systematic diagnostic account of what works, what fails, and why.

**Positive results.** A 0.4M Conv1D encoder pre-trained for 20 epochs on 2 717 unlabelled recordings reaches 0.775 WIN BACC (frozen), matching the published ContraWR result and within 3.9 pp of LaBraM-Base. Corruption augmentation (+time masking + spike injection) is the single most effective ingredient (+0.014 WIN, +0.029 REC). Fine-tuning the pre-trained encoder reaches 0.812 REC BACC (3 seeds, final epoch), matching EEGNet (0.812) with no labels used during pre-training.

**Diagnostic findings.**

1. **VICReg coefficient bug**: inv_coeff=1 vs. the paper-correct 25. The bug under-weighted variance regularisation 25×, causing mild collapse that was invisible from loss curves but harmful to BACC. Fixed and confirmed via EEGPT inv_loss: 0.103 (buggy) → 0.018 (fixed) at epoch 11.
2. **Per-window / per-recording gap**: the framework defaults to REC-WIN ($N$=16), which is ∼5 pp above WIN for the same encoder. Comparing REC to WIN across papers produces a spurious advantage. This caused our initial claim of "beating LaBraM" to be retracted.

3. **EEGPT architectural conflict**: channel-pool (19 ch $\rightarrow$ 1 token) combined with channel-drop augmentation creates cross-view inconsistency that the invariance loss cannot resolve. Ablation jobs pending.

### 5.2 Open Questions (Decision Log)

Six scientific decisions remain unresolved as of this writing:

**D1** Is the EEGPT channel-drop conflict a second cause of collapse (independent of the VICReg bug), or was the bug the whole story? Pending jobs 75752 (fix only) and 75806 (fix + no channel-drop).

**D2** Which variable explains the 3.9 pp gap to LaBraM-Base: tokeniser quality, model scale, or pretraining duration? Each requires a separate 20–40 epoch run; untested.

**D3** Should the probe be LogReg (SSL community standard) or MLP? Our MLP probe inflates absolute numbers relative to published SimCLR comparisons. Switching and re-running all experiments is needed for fair cross-paper comparison.

**D4** Does spectral regularisation add independent signal at the correct VICReg coefficients? The current spec_fixed result (0.836) was trained with the bug; spec + fix combination untested.

**D5** At what per-window BACC threshold is EEGPT "rehabilitated"? Pending jobs 75752/75806.

**D6** Does the mc_transformer (3.65M, multi-corpus, job 75792) change the capacity narrative? If it underperforms the 0.4M Conv, architecture — not capacity — is the binding constraint.

### 5.3 Future Work

1. **Neural tokeniser for LaBraM-style encoder.** LaBraM's pretrained vector-quantised neural spectrum tokeniser is likely the key differentiator at 5.8M params. Training this tokeniser separately (masked patch prediction on Fourier spectra) and plugging into our 1.2M backbone is the highest-priority experiment for closing the 3.9 pp gap.
2. **Longer pre-training.** All runs used 20 epochs. LaBraM used ~200 epochs on 2 500h data. Even doubling to 40 epochs may yield measurable improvement.
3. **EEGPT with per-encoder augmentation policy.** Once D1 is resolved, EEGPT should have a dedicated augmentation stack that does not conflict with channel-pooling. The fix may recover EEGPT to a competitive level.
4. **LogReg probe re-evaluation.** Running all final checkpoints through a scikit-learn LogisticRegression (C=0.316, max_iter=1000) probe would allow direct comparison to SimCLR, MoCo, and BYOL literature without confounding probe strength.
5. **Aperiodic reliance audit.** The spectral audit of Bindra & Panwar [3] showed that EEG deep learning models on TUAB rely significantly on the aperiodic ($1/f$-shaped) broadband envelope ($\Delta$BACC 0.07–0.13 after flattening). Our SSL encoders should be subjected to the same audit to determine whether EB-JEPA representations capture oscillatory biomarkers or merely the broadband slope.
6. **Predictive JEPA completion.** The EMA-based predictive variant (GRU predictor, frame-prediction objective) converged for BIOT but not Conv. Evaluating probe BACC would determine whether predictive objectives yield richer representations for clinical EEG, or merely learn temporal autocorrelation.

### 5.4 Methodological Lessons

This project illustrates four recurring failure modes in EEG deep learning:

1. **Comparison without protocol matching.** WIN vs. REC is the EEG-specific

instance of a general problem: different papers measure different quantities. Always implement the exact same evaluation as the paper you compare against before reporting a gap.

2. **Loss coefficient bugs that are invisible from convergence.** A 25× underweighted loss term may still cause training to converge to a "good enough" minimum; the bug only becomes visible at downstream evaluation. Unit-testing that loss components contribute in the expected ratio is advisable.
3. **Augmentation–architecture coupling.** Augmentation choices that are benign for one architecture can be catastrophic for another (channel-drop + channel-pool). The augmentation stack should be designed or ablated per encoder family, not applied uniformly.
4. **Best-epoch selection peeks at the test set.** Reporting the epoch with the highest *evaluation* accuracy uses the test set for model selection — a subtle leak. Our fine-tuned probe appeared to reach 0.837 REC BACC at its best epoch but only $0.812 \pm 0.004$ at the final epoch (3 seeds); the 2.5 pp gap is pure peeking, and it was exactly the margin that made SSL look like it *beat* EEGNet rather than tying it. Every model must share one fixed stopping rule (we report final-epoch, seed-averaged); selecting per-model best epochs against the same split you report on manufactures wins that vanish under a held-out protocol.

**Code.** All models, training scripts, and evaluation code are available in the project repository. The science-check protocol (`.claude/commands/science-check.md`) enforces pre-registration of falsification criteria before any new experiment.

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